

SMART ATTENDANCE ADVISOR

Intelligent Agent – Micro Level Project

1. Problem Formulation

The Smart Attendance Advisor is an intelligent agent designed to help students maintain the required attendance percentage for each subject.

Students may not know how many classes they can safely skip or how many upcoming classes they must attend when their attendance falls below the required percentage. The system accepts classes attended, classes conducted, and the required threshold, then provides a recommendation.

Agent Goal: Maintain attendance at or above the required threshold.

Inputs: Required attendance percentage, subject name, classes attended, and classes conducted.

Outputs: Current attendance, status, maximum safe skips, or minimum classes that must be attended.

2. Approach

The project is implemented in Python and follows a goal-directed reasoning approach. The agent observes the current attendance state and compares it with its attendance goal.

Attendance Percentage = (Classes Attended / Classes Conducted) × 100

If attendance is equal to or greater than the threshold, the agent calculates the maximum future classes that can be skipped while remaining compliant. If attendance is below the threshold, it calculates the minimum number of consecutive upcoming classes that must be attended to reach the goal.

The implementation also handles zero classes conducted, negative values, attended classes greater than conducted classes, and invalid thresholds.

3. Algorithm

1. Read the required attendance threshold.
2. Read the number of subjects.
3. Read the subject name.
4. Read classes attended and classes conducted.
5. Validate the input values.
6. Calculate the current attendance percentage.
7. Compare attendance with the required threshold.
8. If the goal is satisfied, calculate safe skips.
9. If the goal is not satisfied, calculate required classes to attend.
10. Display the recommendation.
11. Repeat for all subjects.

4. Goal-Directed Decision States

State	Decision
COMPLIANT	Goal satisfied; calculate maximum safe skips.
BELOW_GOAL	Goal not satisfied; calculate classes that must be attended.
NO_DATA	No classes have been conducted yet.
ERROR	Attendance input is invalid.

5. Complexity

Attendance calculation requires constant time: $O(1)$. For the below-threshold case, if k additional classes are required, the loop takes $O(k)$ time. The space complexity is $O(1)$ because no large data structure is required.

6. Edge Case Handling

- Zero classes conducted: avoids division by zero and reports no attendance data.
- Negative values: rejected as invalid.
- Attended greater than conducted: rejected as invalid.
- Exactly 75%: treated as compliant.
- Below 75%: calculates the minimum consecutive classes to attend.
- Invalid threshold: only 1% to 100% is accepted.

7. Results

The Smart Attendance Advisor was implemented and tested successfully.

Test Item	Value
Required attendance	75%
Subject	Chem
Classes attended	23
Classes conducted	24
Current attendance	95.83%
Maximum safe skips	6 classes
Status	Goal satisfied

Calculation: $(23 / 24) \times 100 = 95.83\%$. At a 75% threshold, the student can skip up to 6 future classes and still remain above the required goal.

Automated testing: The project was tested using pytest and **9 tests passed successfully**. The tests cover normal calculations, full attendance, zero classes, below-threshold attendance, exact-threshold attendance, safe skipping, invalid values, and decision cases.

8. Reflection

The main challenge was designing the decision-making logic for both situations: attendance above the threshold and attendance below the threshold. The project demonstrates goal-directed reasoning because the attendance goal directly determines the recommendation.

Testing helped verify that the calculation and decision logic work correctly for normal cases and edge cases. Future improvements could include a graphical user interface, attendance history, report generation, and prediction using future class schedules.

AI tools and online references may have been consulted during development, but the final implementation was tested and verified by the student.

9. Conclusion

The Smart Attendance Advisor successfully demonstrates a simple goal-directed intelligent agent. It calculates subject-wise attendance, compares the current state with a required attendance goal, and provides an actionable recommendation.

The system can tell a student how many classes may be skipped safely or how many upcoming classes must be attended to achieve compliance. The successful automated tests demonstrate that the attendance calculation and decision-making logic work as expected.